PROBLEM 20

Heavy Particle on a Rotating Circumference

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Energy in a Rotating Frame

A heavy particle is moving along a smooth circumference of radius R and center O. One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity ω about it. The position of the particle with respect to the circumference is described by the angle θ between the particle and the lowest point of the circumference.

(a) Discuss the conservation of mechanical energy in the moving reference frame and compute its value.

(b) Compute the reaction of the circumference in the direction normal to its plane, as a function of θ.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m slides on a smooth circular wire of radius R and centre O; the wire turns with constant angular velocity ω about its vertical diameter. The angle θ is measured from the lowest point.
- **Why the moving frame.** In the frame S_1 attached to the wire the constraint is a fixed circle, the motion has one coordinate θ, and the reaction has two clear components (in the plane of the wire, normal to it). The question itself asks for the energy "in the moving reference frame", which requires the inertia forces of Notes Eq. 4.62 and their potentials (Slides 29–31).
- **Hypotheses:**

1. S_0 fixed to the ground is inertial; O is fixed; uniform gravity $-g\,\mathbf{k}$.
2. The wire is smooth: the reaction has no component along the tangent of the wire.
3. ω is constant.

- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $\mathbf{k}_0$ upward along the rotation axis.
- **Moving frame:** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire: $\mathbf{k}_1 = \mathbf{k}_0$, $\mathbf{i}_1$ horizontal in the plane of the wire (on the side where $\theta > 0$), $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ normal to the plane of the wire. The plane of the wire is at azimuth ωt from $\mathbf{i}_0$.
- **Wire basis:** $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$, radial (from O to P), tangent (increasing θ) and normal to the plane; deduced in the next section.
- **Coordinate and degrees of freedom:** the particle is constrained to a curve fixed in S_1: **1 DOF**, θ, the angle of OP measured at O from the downward vertical $-\mathbf{k}_1$, positive towards $+\mathbf{i}_1$.
- **Unknown reaction:** $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$, with N_r in the plane of the wire (along the radius) and N_b normal to that plane.

Symbol	Meaning	SI unit
m	mass of the particle	kg
R	radius of the circumference	m
ω	spin rate of the circumference (constant)	rad/s
g	gravitational acceleration	m/s^2
θ	angle from the lowest point	rad
$N_r,\ N_b$	reaction components (radial in-plane, normal to the plane)	N
$\theta_0,\ \dot\theta_0$	initial angle and angular rate in S_1	rad, rad/s

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

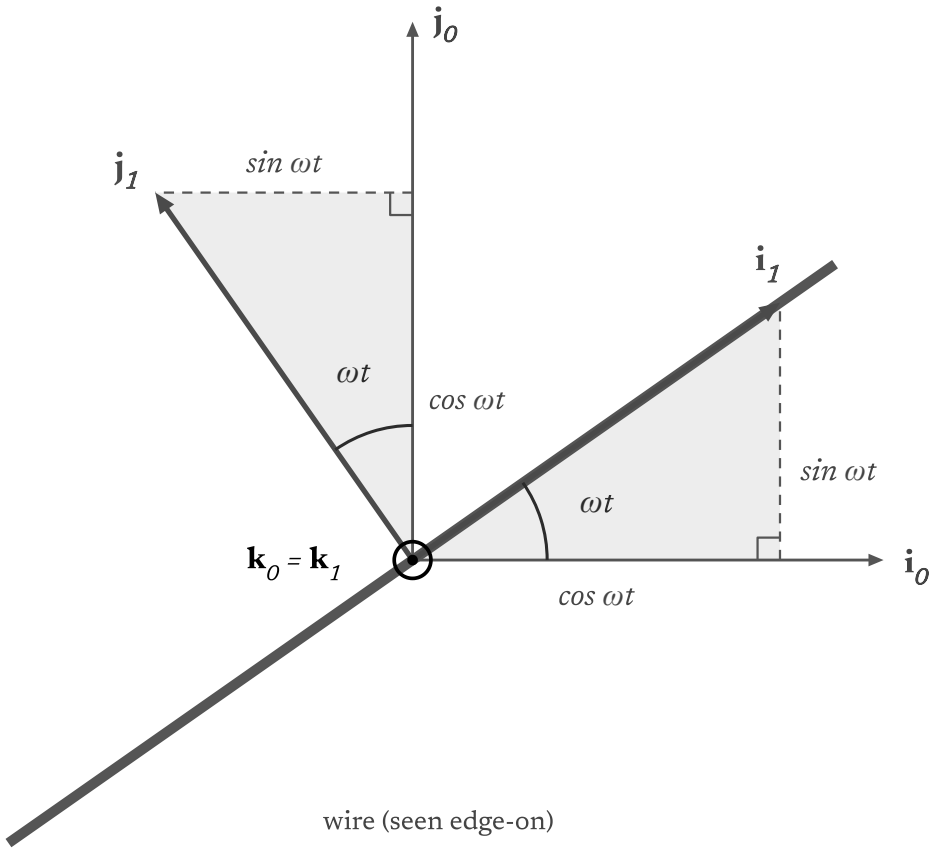


Figure 1. View from $+\mathbf{k}_0$ ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). The wire is seen edge-on, along $\mathbf{i}_1$, turned counter-clockwise by ωt; $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ is normal to it. The opposite leg of $\mathbf{j}_1$ points along $-\mathbf{i}_0$: hence $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down from $+\mathbf{k}_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$), $\mathbf{i}_0$ to the right and, by the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$, $\mathbf{j}_0$ up the page. The plane of the wire is seen edge-on, as a line through O along $\mathbf{i}_1$; it turns counter-clockwise ($\omega > 0$) and makes the angle ωt with $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle with the leg $\cos\omega t$ along $\mathbf{i}_0$ (adjacent) and the leg $\sin\omega t$ along $+\mathbf{j}_0$ (opposite): $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ is $\mathbf{i}_1$ turned 90° counter-clockwise, normal to the trace of the wire. It has turned ωt from $\mathbf{j}_0$, towards the left, so its opposite leg $\sin\omega t$ lies along $-\mathbf{i}_0$: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$ (that is the minus sign of the matrix).
- $\mathbf{k}_1 = \mathbf{k}_0$.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and mutually orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. Poisson check (chain rule): $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1 = \omega\,\mathbf{k}_1 \times \mathbf{i}_1$.

The position vector in the plane of the wire

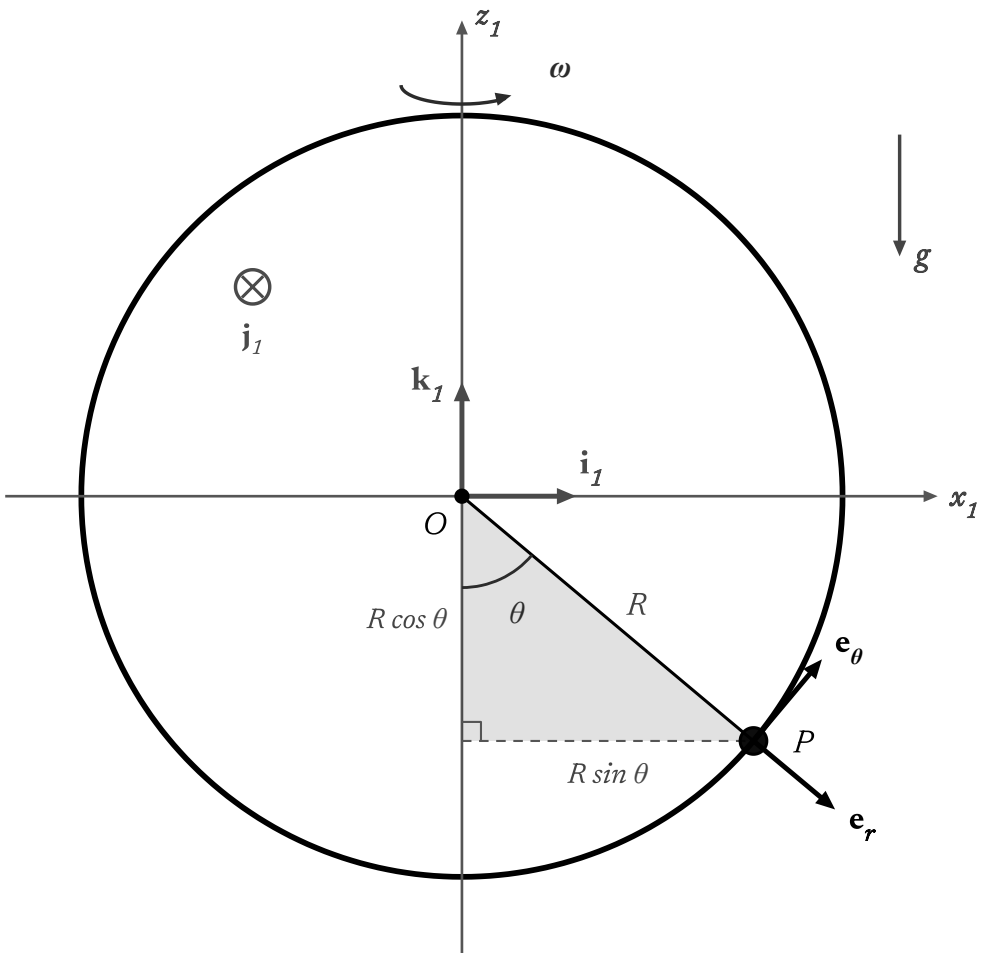


Figure 2. Plane of the wire, fixed in S_1 ($\mathbf{i}_1$ right, $\mathbf{k}_1$ up, so $\mathbf{j}_1$ points into the paper, $\otimes$). In the shaded right triangle OFP the hypotenuse is R, the leg opposite θ is $R\sin\theta$ along $\mathbf{i}_1$ and the adjacent leg is $R\cos\theta$ along $-\mathbf{k}_1$. $\mathbf{e}_r, \mathbf{e}_\theta$ in orange.

The plane of the wire is fixed in S_1 and is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up. Then $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ points into the paper ($\otimes$): fingers from $\mathbf{k}_1$ (up) to $\mathbf{i}_1$ (right), clockwise, so the thumb points away from the reader.

The centre O is the origin, so $\mathbf{r}_1^P = \overrightarrow{OP}$, of length R, making the angle θ with the downward vertical $-\mathbf{k}_1$. Let F be the point of the vertical axis at the height of P. In the shaded right triangle OFP (right angle at F) the hypotenuse is $|OP| = R$; the leg opposite to θ is $FP = R\sin\theta$, horizontal, along $+\mathbf{i}_1$; the leg adjacent to θ is $OF = R\cos\theta$, vertical and pointing down, along $-\mathbf{k}_1$. Hence

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{e}_r = \sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$$

Check: $\theta = 0$ gives $-R\,\mathbf{k}_1$, the lowest point; $\theta = \pi/2$ gives $R\,\mathbf{i}_1$, the end of the horizontal diameter. The part of $\mathbf{r}_1^P$ normal to the axis is the horizontal leg, $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$.

Relation with the standard polar angle. Measured from $\mathbf{i}_1$ towards $\mathbf{k}_1$, the usual polar angle φ gives $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{k}_1$. This matches $\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$ for $\varphi = \theta - \pi/2$: $\cos\left(\theta - \frac{\pi}{2}\right) = \cos\theta\cos\frac{\pi}{2} + \sin\theta\sin\frac{\pi}{2} = \sin\theta$ and $\sin\left(\theta - \frac{\pi}{2}\right) = \sin\theta\cos\frac{\pi}{2} - \cos\theta\sin\frac{\pi}{2} = -\cos\theta$. So θ is the polar angle measured from $-\mathbf{k}_1$, chosen so that $\theta = 0$ is the lowest point, as in the statement.

Wire basis. The tangent is the derivative with respect to θ ($\frac{d}{d\theta}\sin\theta = \cos\theta$, $\frac{d}{d\theta}(-\cos\theta) = \sin\theta$), already of unit length ($\sqrt{\cos^2\theta + \sin^2\theta} = 1$):

$$\mathbf{e}_\theta = \frac{d\mathbf{e}_r}{d\theta} = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1$$

Since $\mathbf{e}_r \times \mathbf{e}_\theta = (-\cos\theta)(\cos\theta)(\mathbf{k}_1 \times \mathbf{i}_1) + (\sin\theta)(\sin\theta)(\mathbf{i}_1 \times \mathbf{k}_1) = -\cos^2\theta\,\mathbf{j}_1 - \sin^2\theta\,\mathbf{j}_1 = -\mathbf{j}_1$ (with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$), the right-handed triad is $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$; $-\mathbf{j}_1$ points out of the paper in the plane view. Its columns in $\mathcal{B}_1$:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_1R_w] = \sin\theta(0\cdot 0 + \sin\theta) - \cos\theta(0\cdot 0 - \cos\theta) = 1$$

and the columns are orthonormal ($\sin^2\theta + \cos^2\theta = 1$, $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so $[{}_1R_w][{}_1R_w]^T = I$.

Derivatives in S_1 (chain rule): $\frac{d\mathbf{e}_r}{dt}\big|_1 = \frac{d\mathbf{e}_r}{d\theta}\dot\theta = \dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = \dot\theta\,\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = \dot\theta\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{e}_r$.

Kinematics

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = R\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^P = R\ddot\theta\,\mathbf{e}_\theta + R\dot\theta\left(-\dot\theta\,\mathbf{e}_r\right) = R\ddot\theta\,\mathbf{e}_\theta - R\dot\theta^2\,\mathbf{e}_r$$

(R is constant, so $\mathbf{v}_1^P = R\,\frac{d\mathbf{e}_r}{dt}\big|_1$; then the product rule on $R\dot\theta\,\mathbf{e}_\theta$: $\frac{d}{dt}(R\dot\theta) = R\ddot\theta$ and $\frac{d\mathbf{e}_\theta}{dt}\big|_1 = -\dot\theta\,\mathbf{e}_r$.)

The absolute velocity (Notes Eq. 3.26), used below, is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = R\dot\theta\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$ (from $\omega\mathbf{k}_1\times(R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1) = R\omega\sin\theta\,\mathbf{j}_1$).

Forces and Equations of Motion

Forces seen from S_1 (Notes Eqs. 4.62–4.63):

- weight $-mg\,\mathbf{k}_1$ and reaction $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$ (actual forces);
- $\mathbf{F}_A = \mathbf{0}$ (O fixed) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$);
- centrifugal force: $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$ (the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$), so (Slide 30)

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = m\omega^2 R\sin\theta\,\mathbf{i}_1$$

Component check: $\omega\mathbf{k}_1\times(R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1) = R\omega\sin\theta\,\mathbf{j}_1$, then $\omega\mathbf{k}_1\times R\omega\sin\theta\,\mathbf{j}_1 = -R\omega^2\sin\theta\,\mathbf{i}_1$, and the minus sign of $\mathbf{F}_C$ gives $+m\omega^2 R\sin\theta\,\mathbf{i}_1$;

- Coriolis force, with $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\omega\mathbf{k}_1\times R\dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = -2m\omega R\dot\theta\cos\theta\,\mathbf{j}_1$$

Equation of motion (projection on $\mathbf{e}_\theta$, where $\mathbf{N}$ has no component). Using $\mathbf{k}_1\cdot\mathbf{e}_\theta = \sin\theta$, $\mathbf{i}_1\cdot\mathbf{e}_\theta = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$:

$$mR\ddot{\theta} = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta \quad \Longrightarrow \quad \ddot{\theta} = -\frac{g}{R}\sin\theta + \omega^2\sin\theta\cos\theta$$

Projection on $\mathbf{j}_1$ (normal to the plane). The relative motion stays in the plane, so $\mathbf{a}_1^P \cdot \mathbf{j}_1 = 0$; the weight and the centrifugal force lie in the plane; only N_b and the Coriolis force remain:

$$m\,\mathbf{a}_1^P \cdot \mathbf{j}_1 = 0 = N_b - 2m\omega R\dot{\theta}\cos\theta \quad \Longrightarrow \quad N_b = 2m\omega R\dot{\theta}\cos\theta$$

Projection on $\mathbf{e}_r$ ($\mathbf{k}_1 \cdot \mathbf{e}_r = -\cos\theta, \mathbf{i}_1 \cdot \mathbf{e}_r = \sin\theta$):

$$-mR\dot{\theta}^2 = N_r + mg\cos\theta + m\omega^2 R\sin^2\theta \quad \Longrightarrow \quad N_r = -mR\dot{\theta}^2 - mg\cos\theta - m\omega^2 R\sin^2\theta$$

Answers and Checks

(a) Mechanical energy in the moving frame

- **Justification:** E_1^P is conserved when every force that does work in S_1 derives from a potential (Slide 29). We check the forces one by one using their power in S_1, $\mathbf{F} \cdot \mathbf{v}_1^P$.
- **Reaction:** $\mathbf{N} \cdot \mathbf{v}_1^P = R\dot{\theta}\,(N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1) \cdot \mathbf{e}_\theta = 0$: the wire is smooth and **fixed in** S_1, so it does no work in S_1.
- **Coriolis force:** $\mathbf{F}_{COR} \cdot \mathbf{v}_1^P = 0$ always (Slide 31); here explicitly $\mathbf{j}_1 \cdot \mathbf{e}_\theta = 0$.
- **Euler force and origin force:** both zero.
- **Weight:** conservative, $V_g = mgz_1 = -mgR\cos\theta$ (zero at the height of O).
- **Centrifugal force:** $\boldsymbol{\omega}_{10}$ is constant, so it has the potential (Slide 30) $V_1 = -\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 R^2\sin^2\theta$ (zero on the axis).

Therefore E_1^P **is conserved**:

$$\boxed{E_1^P = \frac{1}{2}mR^2\dot{\theta}^2 - mgR\cos\theta - \frac{1}{2}m\omega^2 R^2\sin^2\theta = \frac{1}{2}mR^2\dot{\theta}_0^2 - mgR\cos\theta_0 - \frac{1}{2}m\omega^2 R^2\sin^2\theta_0}$$

its value being fixed by the initial conditions $\theta(0)=\theta_0$, $\dot\theta(0)=\dot\theta_0$ (the statement gives none, so the constant is left in terms of them).

Check by differentiation (chain rule on each term, then the EOM):

$$\frac{dE_1^P}{dt} = mR^2\dot\theta\ddot\theta + mgR\sin\theta\,\dot\theta - m\omega^2R^2\sin\theta\cos\theta\,\dot\theta = mR^2\dot\theta\left[\ddot\theta + \frac{g}{R}\sin\theta - \omega^2\sin\theta\cos\theta\right] = 0$$

The energy in S_0 is not conserved. In S_0 the wire moves, and its reaction does work: with $\mathbf{v}_0^P = R\dot\theta\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$, the power of $\mathbf{N}$ in S_0 is $\mathbf{N}\cdot\mathbf{v}_0^P = N_b\,\omega R\sin\theta \neq 0$. This is why the moving frame is the natural one for an energy integral here.

(b) Reaction normal to the plane of the circumference

> **Pedagogical Justification:** the projection of Notes Eq. 4.63 on $\mathbf{j}_1$ gave $N_b = 2m\omega R\dot\theta\cos\theta$; to write it as a function of θ we eliminate $\dot\theta$ with the energy integral of part (a).

Solving the energy integral for $R\dot\theta$:

$$R^2\dot\theta^2 = \frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta \quad\Longrightarrow\quad R\dot\theta = \pm\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}$$

$$\boxed{N_b(\theta) = \pm\,2m\omega\cos\theta\,\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}}$$

with the sign of $\dot\theta$ (positive when θ increases) and E_1^P from part (a). For completeness, the in-plane component is $N_r = -mR\dot\theta^2 - mg\cos\theta - m\omega^2R\sin^2\theta$.

Checks

- **Origin of N_b.** The normal-to-plane reaction is entirely due to the Coriolis force: the wire must push the particle sideways so that it keeps turning with the plane while its distance to the axis changes. It vanishes when $\dot\theta = 0$ and when $\cos\theta = 0$ (the particle crosses the horizontal diameter, where $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).
- **Relative equilibria** (extension, from $\ddot\theta = 0$ with $\dot\theta = 0$): $\sin\theta\left(\omega^2\cos\theta - g/R\right) = 0$, i.e. $\theta = 0, \theta = \pi$, and, if $\omega^2R > g$, $\cos\theta = g/(\omega^2R)$; at all of them $N_b = 0$.

- **Limit $\omega \to 0$:** $N_b \to 0$ and $E_1^P \to \tfrac{1}{2}mR^2\dot\theta^2 - mgR\cos\theta$, the energy of the simple pendulum.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin\theta\cos\theta - \cos\theta\sin\theta = 0$, and both are normal to $\mathbf{j}_1$.
- **Dimensions:** each term of E_1^P is $\mathrm{kg\,m^2\,s^{-2}} = \mathrm{J}$; $[N_b] = [m\omega R\dot\theta] = \mathrm{kg\,s^{-1}\,m\,s^{-1}} = \mathrm{N}$; inside the root, $[2E_1^P/m] = [gR] = [\omega^2 R^2] = \mathrm{m^2\,s^{-2}}$.
- **Verification (sympy):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k} + \mathbf{N}$ in S_0 returns the same EOM, N_b and N_r; $dE_1^P/dt = 0$ along the EOM, while dE_0^P/dt equals the power $N_b\,\omega R\sin\theta$ of the reaction in S_0.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 20 in its solution keys, so the results are verified by independent derivation (the sympy check above).

Source: Problems.pdf (PDF page 10); no official key